

Semester	T.E. Semester VI – CMPN
Subject	DevOps Lab
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Laboratory	

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Experiment Number	06
Experiment Title	Setup and Run Selenium Tests in Jenkins Using Maven
Objectives (Skill Set / Knowledge Tested / Imparted)	The primary objective of this experiment is to automate software quality assurance by integrating Selenium automated UI tests into a Jenkins Continuous Integration (CI) pipeline using Maven . It aims to teach developers and QA engineers how to transition tests from a local manual execution environment to an automated, headless browser environment triggers on demand.
Description	This lab experiment guides users through a comprehensive, step-by-step setup divided into seven structured phases. It begins with installing core prerequisites like Java JDK 21 , Maven , and Google Chrome , followed by building a Maven project integrated with JUnit 5 and Selenium WebDriver . The tests are intentionally configured to run in headless mode to ensure compatibility with server environments that lack a graphical user interface. Finally, the project is wired into a Jenkins Freestyle job , where execution is automated and JUnit XML test reports are generated and published directly to the Jenkins dashboard for clear tracking of build success.
Program	Creating the Maven Selenium Project and running Selenium using Maven first :

```

Command Prompt

- surefire-reports
  TEST-tests.ExampleTest.xml
  tests.ExampleTest.txt
- test-classes
  tests
    ExampleTest.class

C:\SeleniumJenkinsDemo>notepad pom.xml
C:\SeleniumJenkinsDemo>notepad src\test\java\tests\ExampleTest.java
C:\SeleniumJenkinsDemo>notepad src\test\java\tests\ExampleTest.java
C:\SeleniumJenkinsDemo>cd C:\SeleniumJenkinsDemo
C:\SeleniumJenkinsDemo>dir
Volume in drive C is OS
Volume Serial Number is D8C5-D4C2

Directory of C:\SeleniumJenkinsDemo

27-08-2026 12:03 <DIR>
27-08-2026 11:54 <DIR> 1,484 pom.xml
27-08-2026 11:45 <DIR> src
27-08-2026 11:55 <DIR> target
1 File(s) 1,484 bytes
3 Dir(s) 339,548,237,824 bytes free

C:\SeleniumJenkinsDemo>run clean test
[INFO] Scanning for projects...
[INFO] -----< com.example.SeleniumJenkinsDemo >-----
[INFO] Building SeleniumJenkinsDemo 1.0-SNAPSHOT

```

Starting Jenkins:

```

Command Prompt

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kirti>cd C:\

C:\>mkdir SeleniumJenkinsDemo\src\test\java\tests
A subdirectory or file SeleniumJenkinsDemo\src\test\java\tests already exists.

C:\>cd SeleniumJenkinsDemo
C:\SeleniumJenkinsDemo>tree /F
Folder PATH listing for volume OS
Volume serial number is 00000062 D8C5:D4C2
C:.
| pom.xml
| src
| | test
| | | java
| | | | tests
| | | | ExampleTest.java
| target
| | generated-test-sources
| | | test-annotations
| | maven-status
| | | maven-compiler-plugin
| | | | testCompile
| | | | | default-testCompile
| | | | | createdFiles.lst
| | | | | inputFiles.lst
| surefire-reports
| | TEST-tests.ExampleTest.xml
| | tests.ExampleTest.txt

```

```

[INFO] --- com.example.selenium:junit5demo:1.0-SNAPSHOT ---
[INFO] Building selenium:junit5demo 1.0-SNAPSHOT
[INFO] Writing pom.xml
[INFO] ---
[INFO] --- clean:1.2.0-clean (default-clean) @ selenium:junit5demo ---
[INFO] Deleting C:\SeleniumJenkinsDemo\target
[INFO] --- resources:3.0.0-resources (default-resources) @ selenium:junit5demo ---
[INFO] skip non existing resourceDirectory C:\SeleniumJenkinsDemo\src\main\resources
[INFO] --- compiler:3.10.0-compile (default-compile) @ selenium:junit5demo ---
[INFO] No sources to compile
[INFO] --- resources:3.0.0-testResources (default-testResources) @ selenium:junit5demo ---
[INFO] skip non existing resourceDirectory C:\SeleniumJenkinsDemo\src\test\resources
[INFO] --- compiler:3.10.0-testCompile (default-testCompile) @ selenium:junit5demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 21] to target\test-classes
[INFO] --- surefire:3.1.1-test (default-test) @ selenium:junit5demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnit4PlatformProvider
[INFO]
[INFO]
[INFO]
[INFO] Running tests ExampleTest
Aug 27, 2026 12:04:09 PM org.openqa.selenium.devtools.CdpVersionFinder findMatchForCdpVersion
WARNING: Unable to find CDP implementation matching 102
Aug 27, 2026 12:04:09 PM org.openqa.selenium.chrome.ChromeDriver <init>
WARNING: Unable to find version of CDP to use for 102.0.7977.41. You may need to include a dependency on a specific version of the CDP using com
stacking similar to 'org.seleniumhq.selenium:selenium-devtools-v102:1.0.0', where the version ("102") matches the version of the chrome-based bro
wser you're using and the version number of the artifact is the same as Selenium's.
Page title = Example Domain
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.833 s -- in tests.ExampleTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 8.727 s
[INFO] Finished at: 2026-08-27T12:04:51+05:30
[INFO]
C:\SeleniumJenkinsDemo>

```

Configure Java and Maven inside Jenkins:



Build Steps

Automate your build process with various tasks like code compilation, testing, and deployment...

Invoke top-level Maven targets

Maven Version

Maven3

Goals

clean test

Advanced

+ Add build step

Maven Installation

+ Add Maven

Maven

Name

Maven3

Maven URL

C:\Program Files\Java\jdk-11.0.10\bin

Install automatically

+ Add Maven

Save

Apply

Selected Maven Test

```

[INFO] Running io.github.kanigalabs:
Aug 27, 2024 12:29:45 PM org.apache.maven.plugins.maven.compiler.plugin:
WARNING: Unable to find the implementation matching the
Aug 27, 2024 12:29:45 PM org.apache.maven.plugins.maven.compiler.plugin:
WARNING: Unable to find version of the to use for the 11.0.10.04. You may need to include a dependency on a specific version of the compiler
something similar to org.apache.maven.plugins:maven-compiler-plugin:4.0.0-alpha-1 where the version "4.0.0-alpha-1" matches the version of the compiler
based version you're using and the version number of the artifact is the same as Selenium's.
App Title : Sample Demo
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s -> io.github.kanigalabs
[INFO]
[INFO]
[INFO]
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO]
[INFO] Total time: 02:23 min
[INFO] Finished at: 2024-08-27T12:30:44+05:30
[INFO]
Finished: SUCCESS

```

